# Introduction to Huawei Cloud HiLens Service

# Foreword

- Huawei HiLens is a multimodal AI development platform featuring a device-cloud synergy. It provides an easy-to-use framework, easy-to-setup development environments, a cloud-based management console, and an AI skill market (a skill can be seen as an AI application ready to run on a camera or another end user device, and it consists of a model and logic code). With Huawei HiLens, you can easily develop and deploy visual and auditory AI applications online, and manage a multitude of connected compute devices.
- This chapter introduces you to the features, advantages, and applications of Huawei HiLens.

# Objectives

- Upon completion of this course, you will be familiar with:
  - The architecture and key features of HiLens.
  - The use cases of HiLens.



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1. **Challenges to Device-Edge-Cloud Synergy**
2. Huawei HiLens Service
3. Huawei HiLens Use Cases




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# Challenges Faced by Developers of End-to-End AI Applications/Solutions

**Smart City**

License plate recognition

Vehicle flow statistics

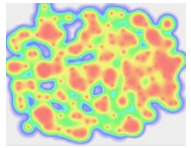
**Smart Campus**

FaceID

Intrusion detection

**Smart Store**

Foot traffic



People flow heatmap

**Examples**

- Difficult to set up the development environment
- High level of expertise required
- Time-consuming development and debugging

- Expensive hardware for end-to-end verification
- High adaptation costs when a large number of devices are used
- Takes long to develop an AI application from scratch

- Complex deployment in a live environment
- Professionals needed

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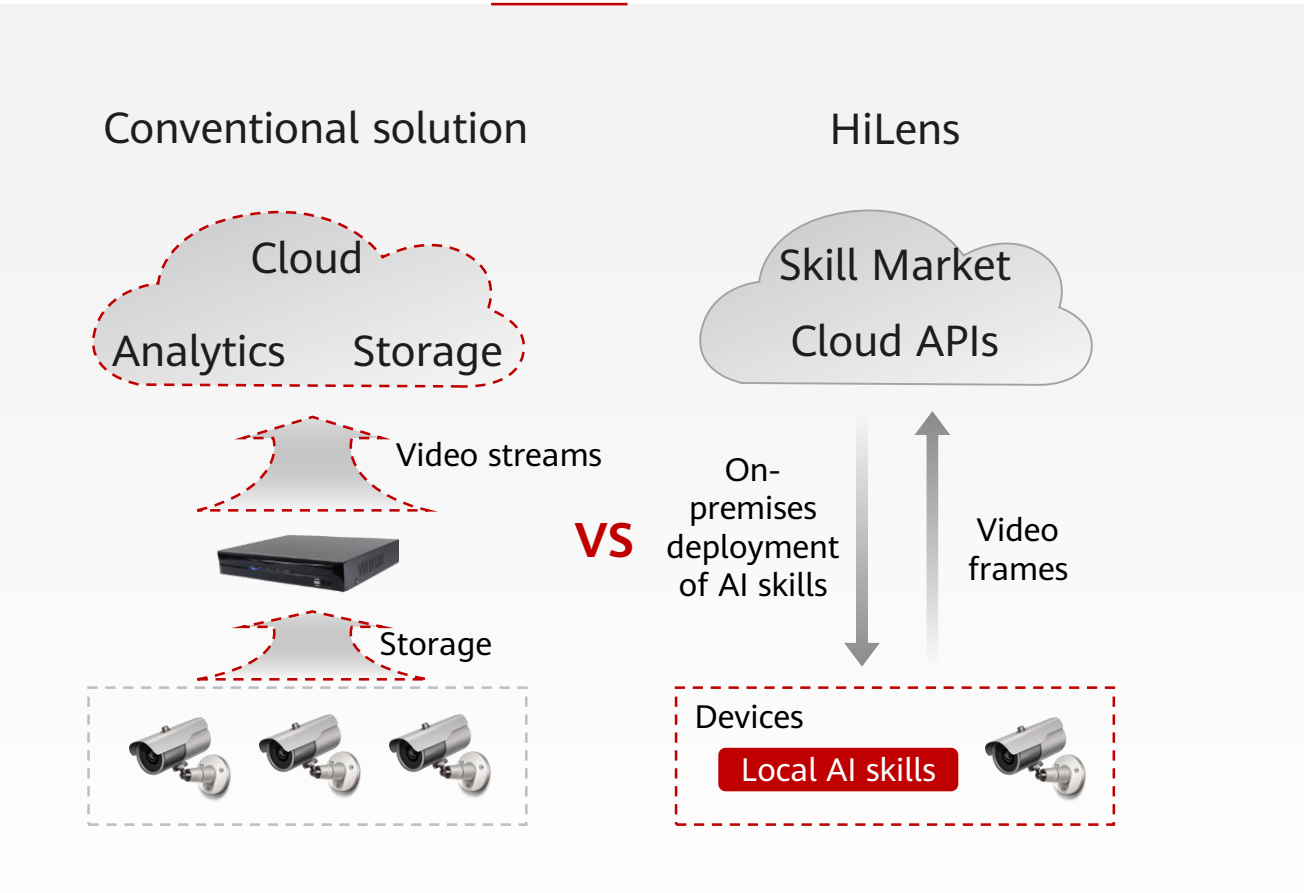
# Advantages of AI with Device-Edge-Cloud Synergy

## A brief comparison




### Lower operating costs

A conventional solution consumes large amounts of **bandwidth, storage, and computing** resources.
HiLens can run on customer premises, which is less costly.

### Better privacy protection

Conventional solution: **all video data** on the cloud
Edge computing with HiLens: no need to transmit data out of the campus

### Faster development

Easy access to many AI applications on the cloud

### Rich application ecosystem

Fast 3rd-party skill development, easy access



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# Device-Edge-Cloud Synergy

- What's edge computing
  - With edge computing, data is processed and stored at the edge of the network to support real-time services and intelligent applications.
- What's intelligent edge
  - Intelligent edge is about deploying AI to edge nodes for intelligent, local data processing.
- What's device-edge-cloud synergy
  - Device-edge-cloud synergy requires a unified technology ecosystem across both cloud and on-premises scenarios, a unified AI architecture, and streamlined AI data flows across the cloud, edge, and end devices, supporting more intelligent edge and end devices.



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# Huawei Cloud HiLens: Agile Development and Deployment of AI Applications with Device-Edge-Cloud Synergy

## Develop once, deploy on various devices, with agile debugging and testing

**Develop on cloud**

**Huawei Cloud**

- Multimodal AI applications (processes/containers) — **Video/image/speech/other AI applications**
- Cloud management platform
- ModelBox, AI application development framework (**open source**)
- AI development plug-ins/tools
- Huawei Cloud ModelArts — **Data** | **Algorithms** | **Models**

Release →
← Buy

Huawei Cloud AI Gallery/Marketplace

**Device-edge-cloud synergy**

**Deploy remotely**

**Edge/Devices** HiLens Agent Inside

| Developer hardware | Smart devices/lite edge nodes | | | | | Smart cameras | | | Consumer devices |
|---|---|---|---|---|---|---|---|---|---|
| HiLens Kit | Atlas500 | IVS 1800 | Atlas800 | Bozz EC02 (RK3399) | 3rd-party edge inference | SDC | Smart mobility devices | 3rd-party smart cameras | |
| HiHope AI development board (RK3568/3588) | | | | | | | | | |
| Ascend 310 + Linux/Ubuntu | Ascend 310 + Linux | | | | X86+GPU | Linux/LiteOS + HiSilicon 35xx | Android + 3rd-party intelligent chips (Adaptation needed) | | Linux/LiteOS /Harmony OS+ HiSilicon 35xx chips |

**Intelligent solutions**

Industrial | Transport | Campus | Finance | City | ...



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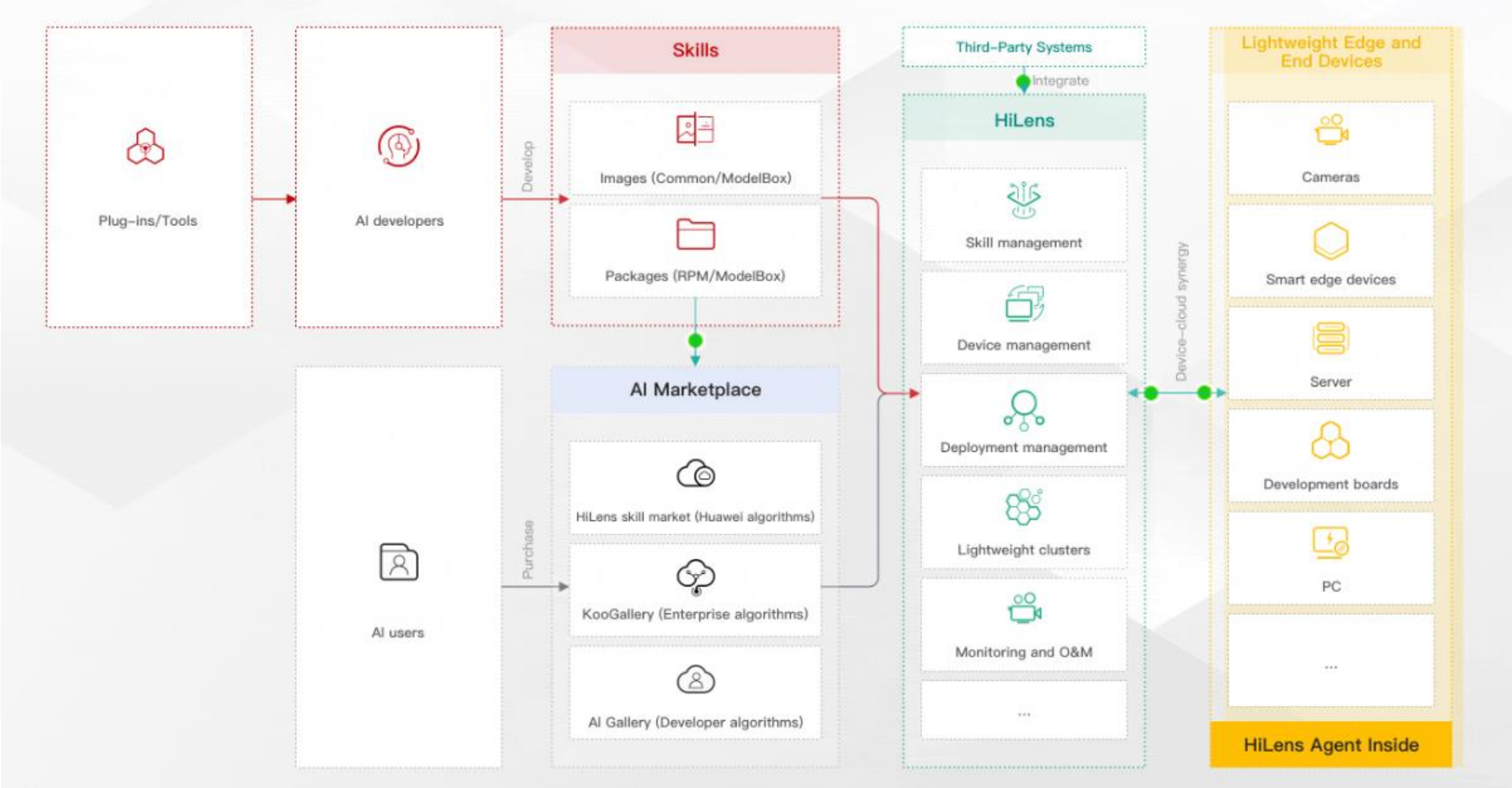
# Device-Cloud Synergy: Flexible Control on the Cloud and Collaboration Across the Cloud and End Devices

Online deployment of AI applications, and real-time running on end devices; further verification and flexible control on the cloud, supporting lightweight application needs.



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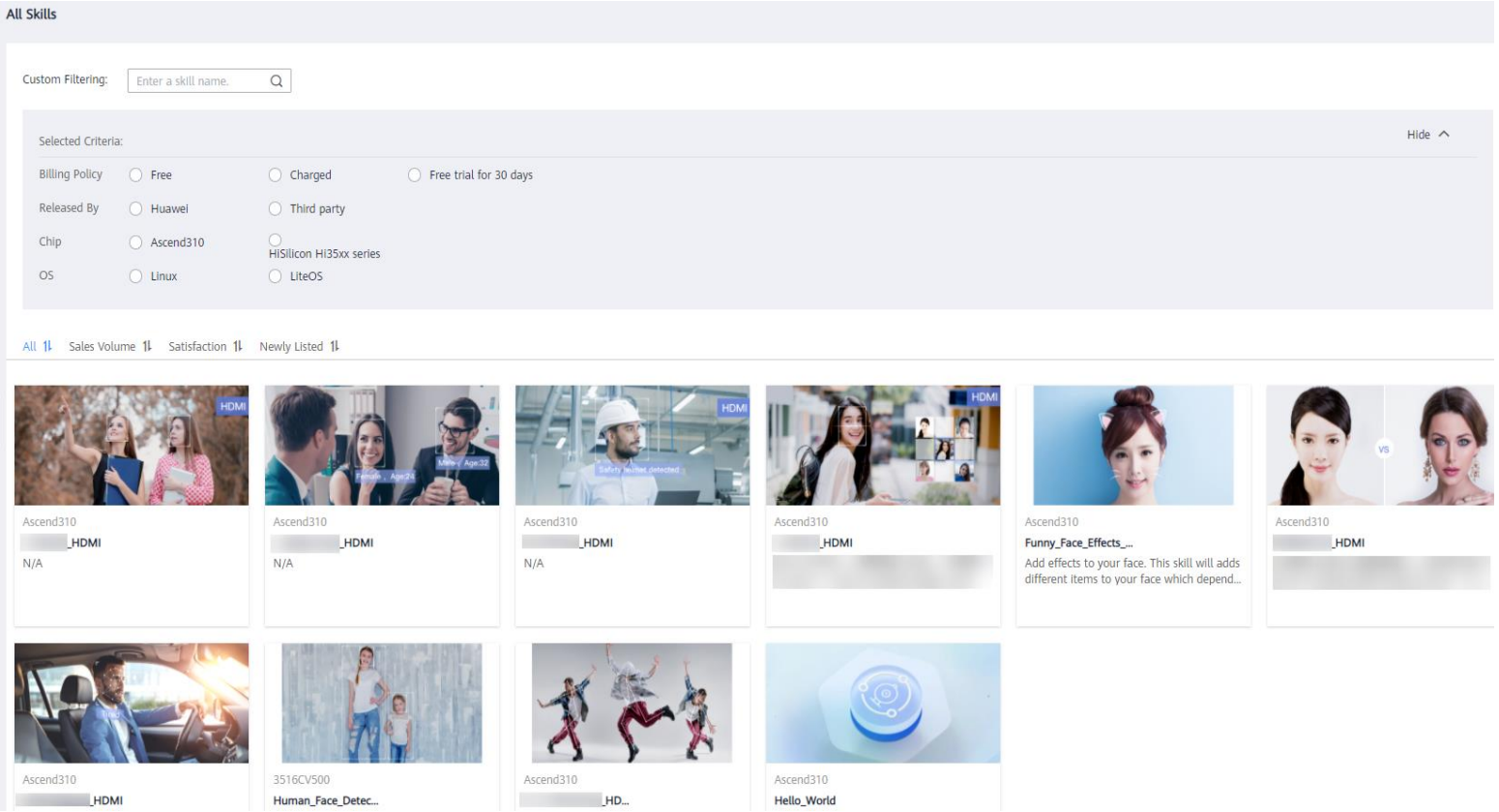
# Quick Application Development and Deployment with Pre-built AI Skills from the HiLens Skill Market

**Skill Market**




**Skill templates (based on HiLens Framework)**

**Select a skill development template**

**Pre-built model and inference code**

One-click installation

**Real-time recognition on end devices**

HiLens Kit



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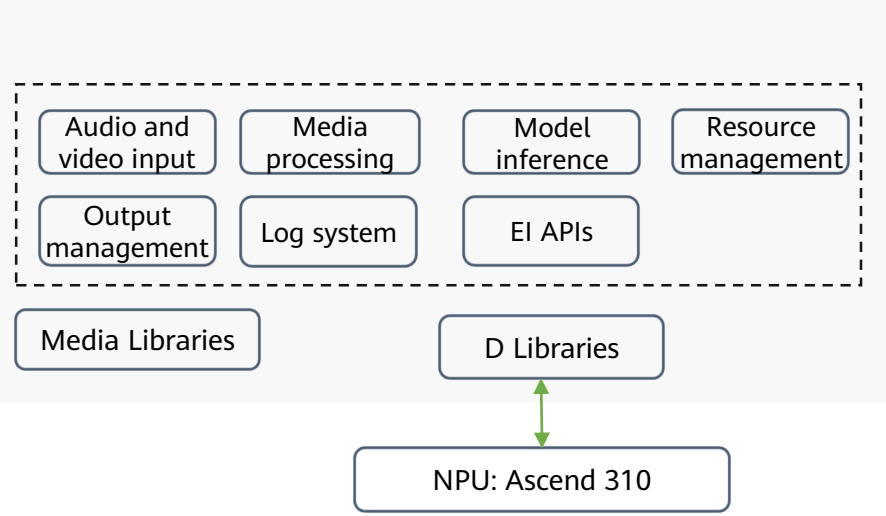
# HiLens Framework + AI Application Runtime Framework ModelBox (1)

HiLens Framework, the AI application development framework, is fully integrated with ModelBox.

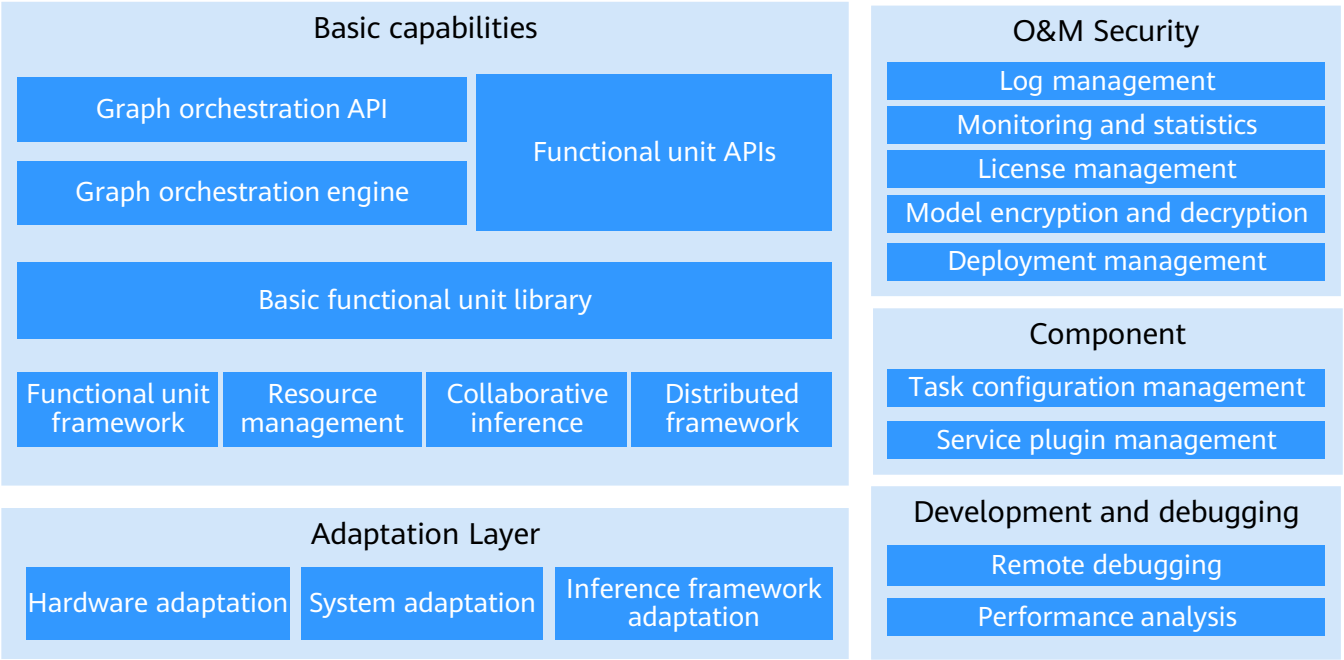
HiLens Framework is an AI inference framework that provides audio and video processing, with a device-cloud synergy. It helps developers easily deploy AI applications to Ascend devices.

**ModelBox provides a unified AI application development and runtime framework featuring device-edge-cloud synergy, helping developers quickly develop audio, video, NLP, and more AI applications and deploy these applications on the cloud, edge, and end devices.**

## HiLens Framework




## ModelBox Architecture



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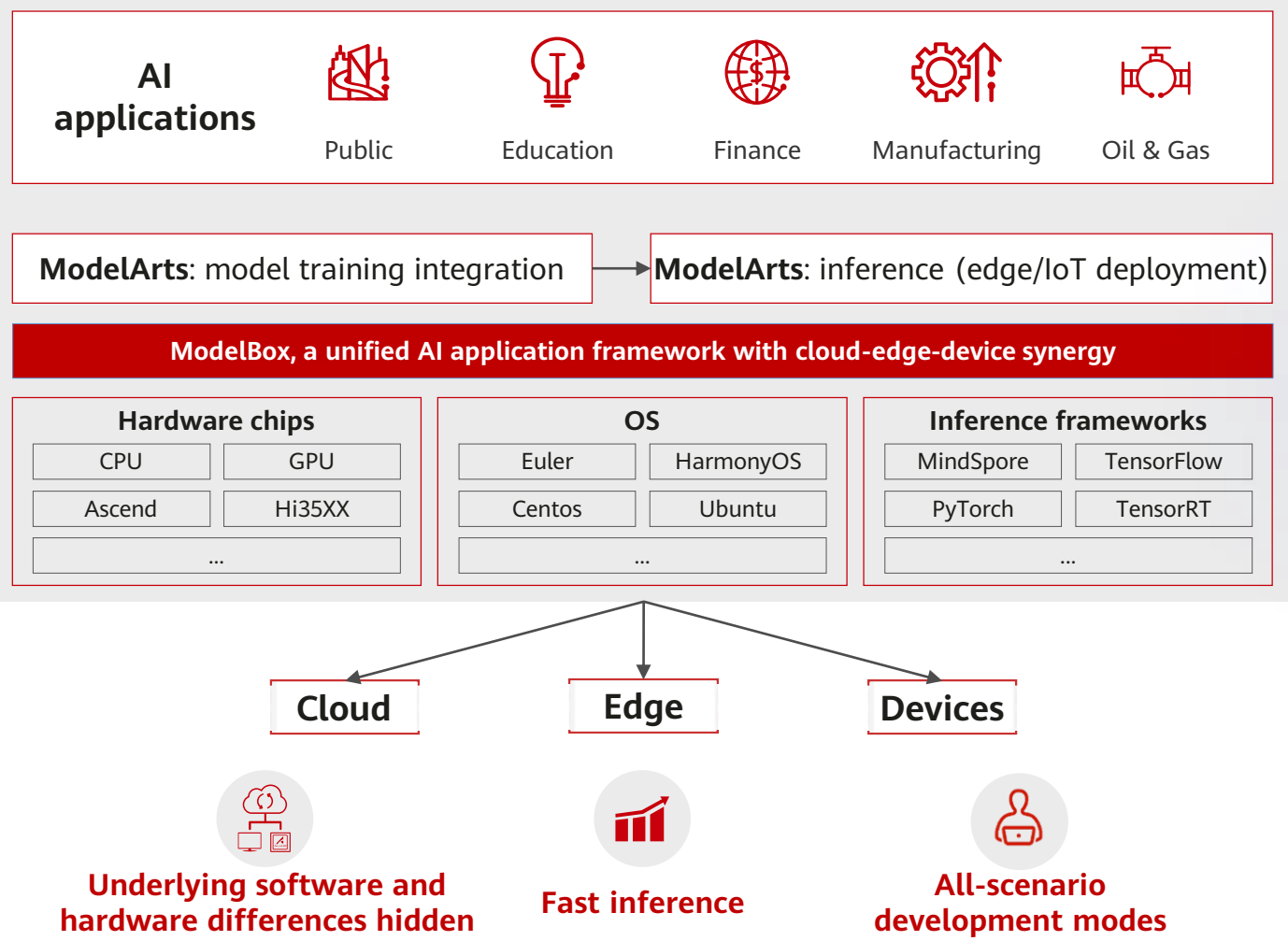
# HiLens Framework + AI Application Runtime Framework ModelBox (2)

Unified application development, deployment, and running across the cloud, edge, and end devices




### Device-edge-cloud joint development and deployment

- Joint development and debugging of AI models/applications across the cloud, edge, and end devices
- Real-time model performance management

### A variety of deployment modes

- Online services: high throughput, low latency, and auto scaling
- Batch inference: efficient resource utilization and distributed compute
- Edge inference: software-hardware integrated solution
- On-device inference: model compression and conversion, lightweight deployment management, support for a wide range of end devices

ModelBox open source home page:

https://modelbox-ai.com/



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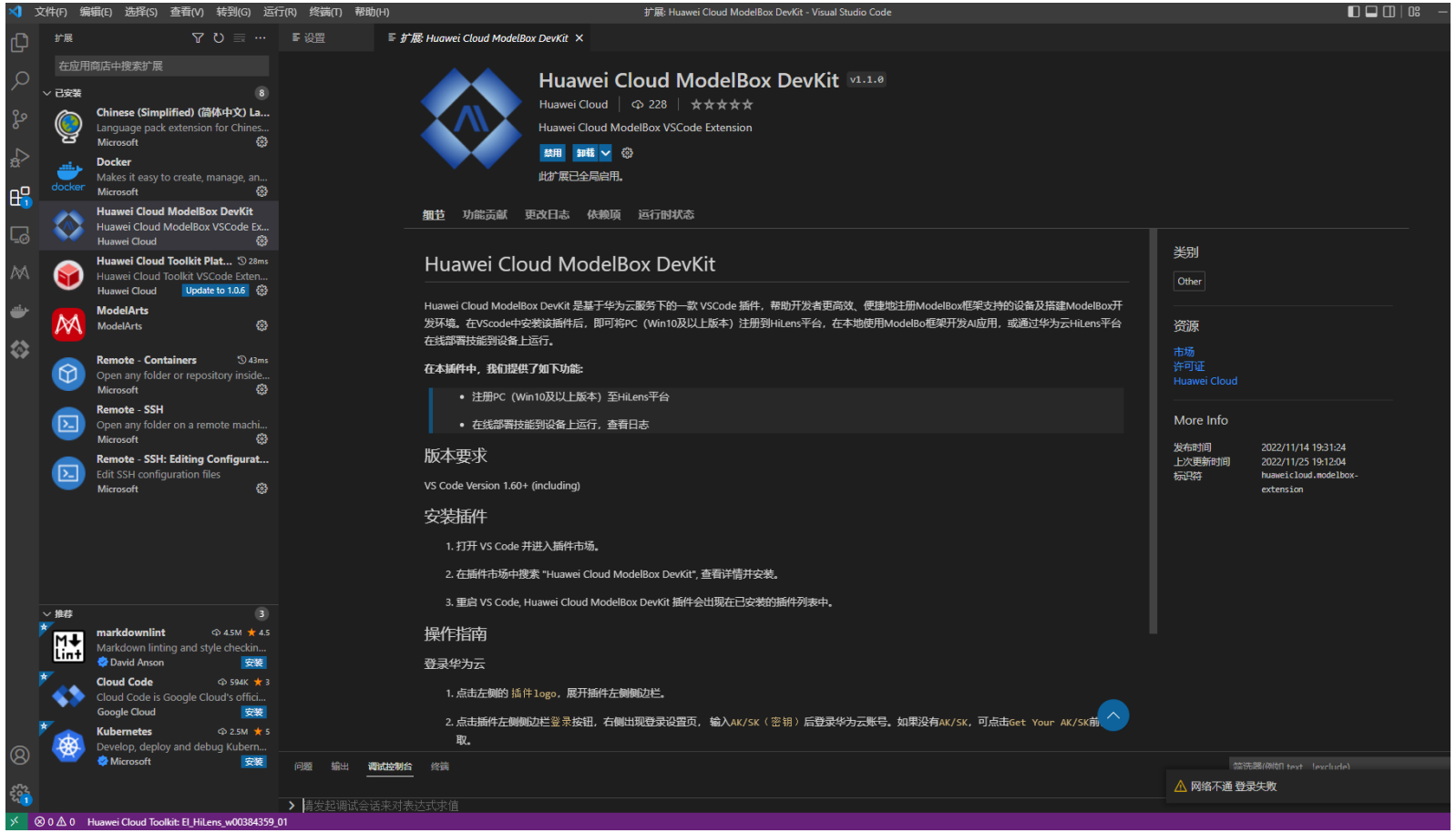
# Efficient Development and Deployment with ModelBox Plugins/Tools

ModelBox SDKs lower the barriers to entry for developers, who can access more resources in the open-source community. The ModelBox VSCode plugin improves developer experience.



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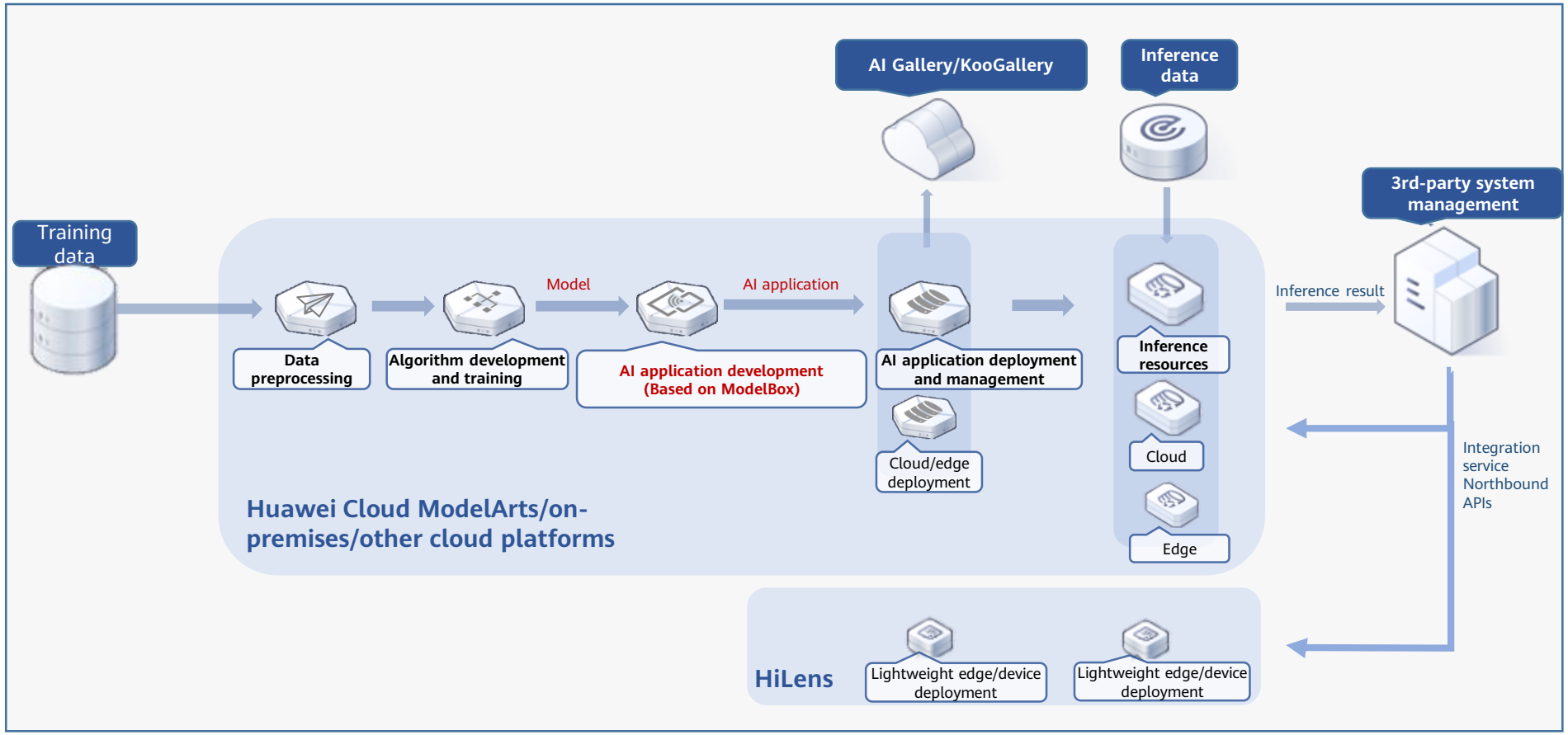
# ModelBox: Joint Development with Device-Edge-Cloud Synergy; Train on Cloud, Deploy Anywhere; and Real-Time Model Update

- Unified algorithm/model packaging formats with ModelBox, same APIs across the cloud, edge, and devices
- A seamless AI marketplace with HiLens for seamless deployment and a robust AI ecosystem



# Huawei HiLens Advantages

### 1. Device-cloud synergy for inference

Devices analyze the collected data locally, reducing the costs of transmitting data to the cloud.

### 2. Unified skill development framework

An easy-to-use skill development framework simplifies development.

### 3. ModelBox

Unified application development, deployment, and running across the cloud, edge, and end devices.

**HiLens**

### 4. Cross-platform design

Huawei-developed Ascend and HiSilicon chips, and other popular chips are supported, covering the needs of mainstream monitoring scenarios.

### 5. An abundance pre-built AI skills

The skill market provides a rich selection of pre-built AI skills for all kinds of applications. Users can quickly deploy these skills onto devices without any development work.

### 6. Developer community

After developing new skills, developers can share them as templates or release them to the skill market, where other users can buy them.



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# HiLens: Multimodal AI Development Application Platform Featuring Device-Cloud Synergy

**HiLens AI-enabled devices**

**A powerful platform with a multimodal AI kernel**

**Unified APIs for inference and computing**

Drone

Home camera

Home bot (babysitting and early education)

Autonomous vehicle

Vehicle-mounted devices

**Huawei Cloud HiLens**

One-stop platform for AI application development, distribution, deployment, and management, skill development, a skill market, and device management, and data management

**Dynamic scheduling of inference resources across the cloud and end devices**

**Dynamic model tuning**

**Device-cloud synergy and low latency**

**Lower costs and higher efficiency**

HiLens deep learning camera Multimodal development kit

Atlas edge station

AI appliance

Ascend 310
An AI SoC with ultra-high computing and energy efficiency

HiSilicon Hi35XX series



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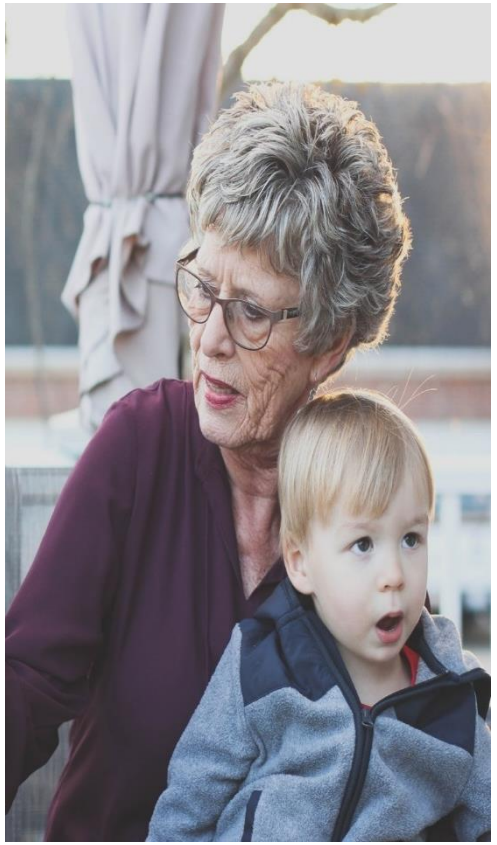
# Home + AI, Making Homes Safer and More Comfortable

**Human figure detection & crying recognition**



Intelligent care for elderly and children

**Human figure detection & facial recognition**

Stranger monitoring and alarms

**Face tagging & attribute recognition from video**

Intelligent family member analysis and retrieval
Children's time album

**Object detection**

Home object & food recognition



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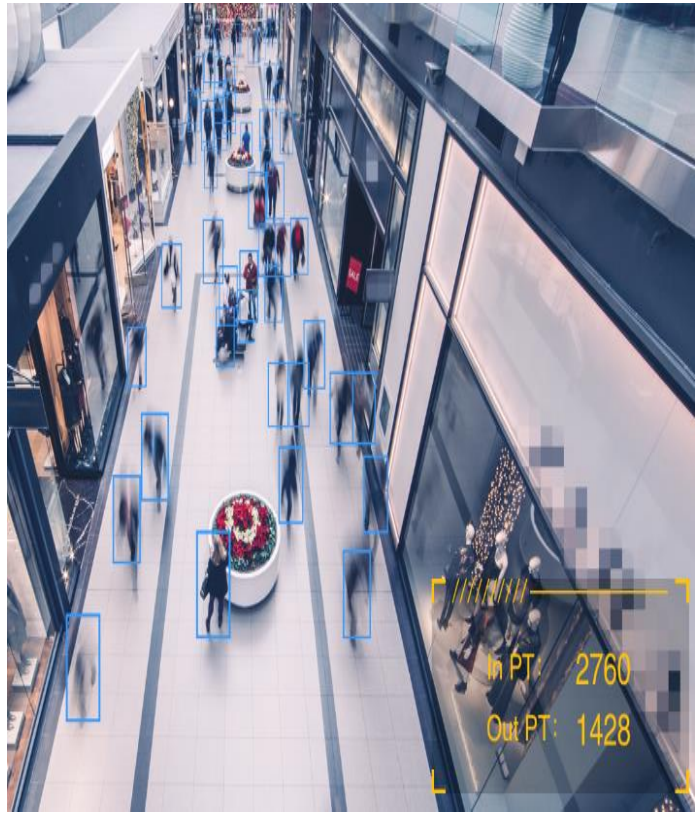
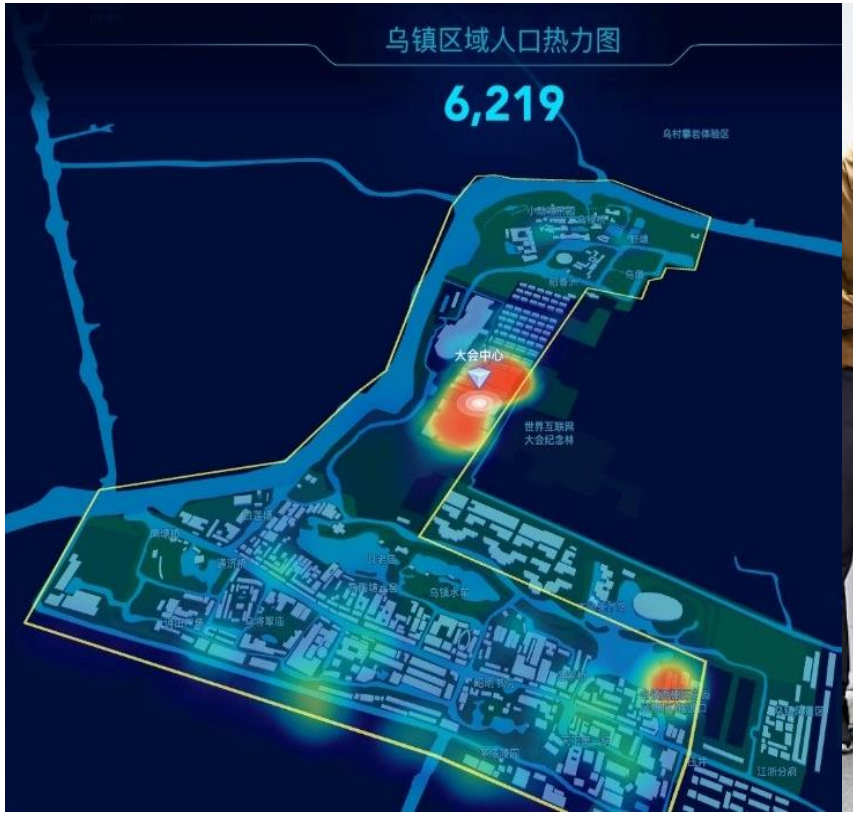
# Shopping Mall + AI, Empowering Intelligent Monitoring and Automated Analysis

**Customer foot traffic**




Counting people entering or leaving a specific area

**Crowd heat map**




Real-time display of crowd density

**Statistics on old and new customers**

Based on facial recognition



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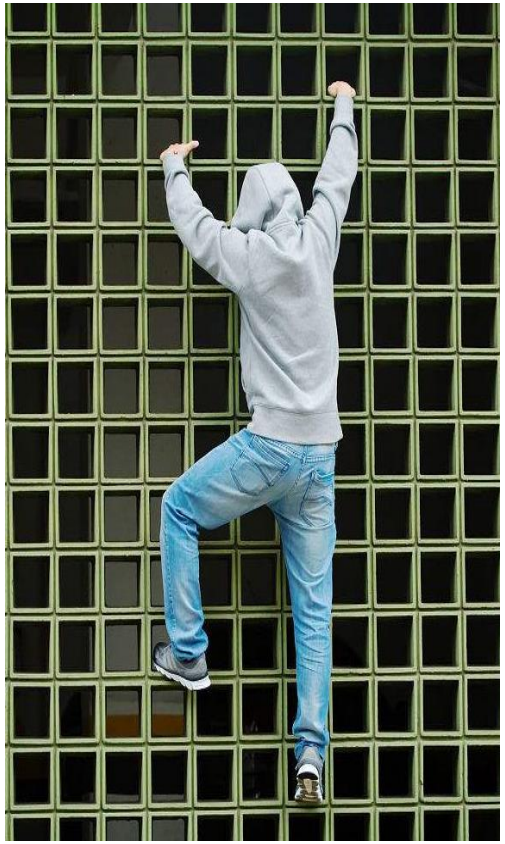
# Campus + AI, Higher Security and Smarter Management

**Intrusion detection**

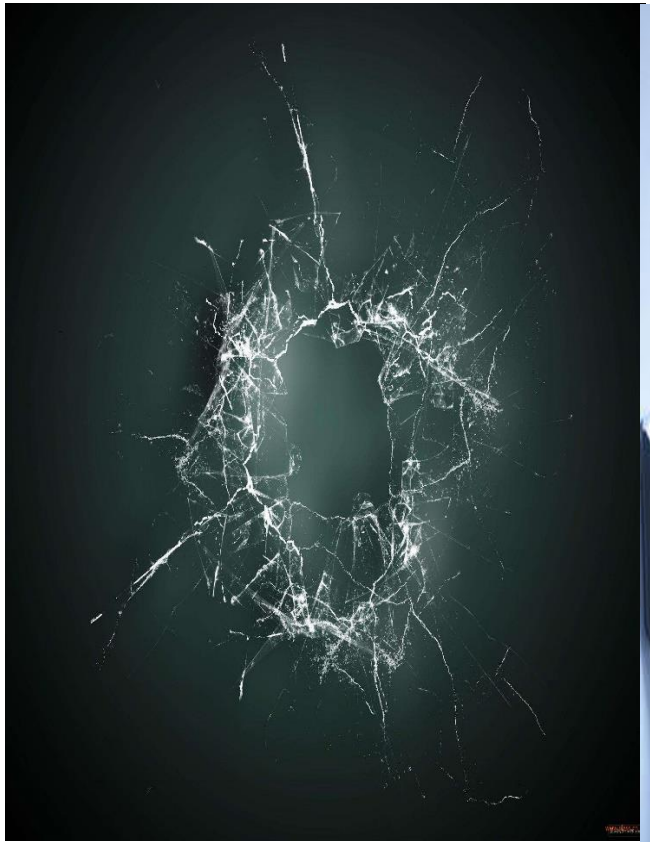


Real-time campus monitoring

**License plate recognition**

Vehicle access control

**Abnormal sound detection**



Alarms for abnormal sounds (e.g. broken glass)

**Access control based on facial recognition**

Intelligent facial recognition



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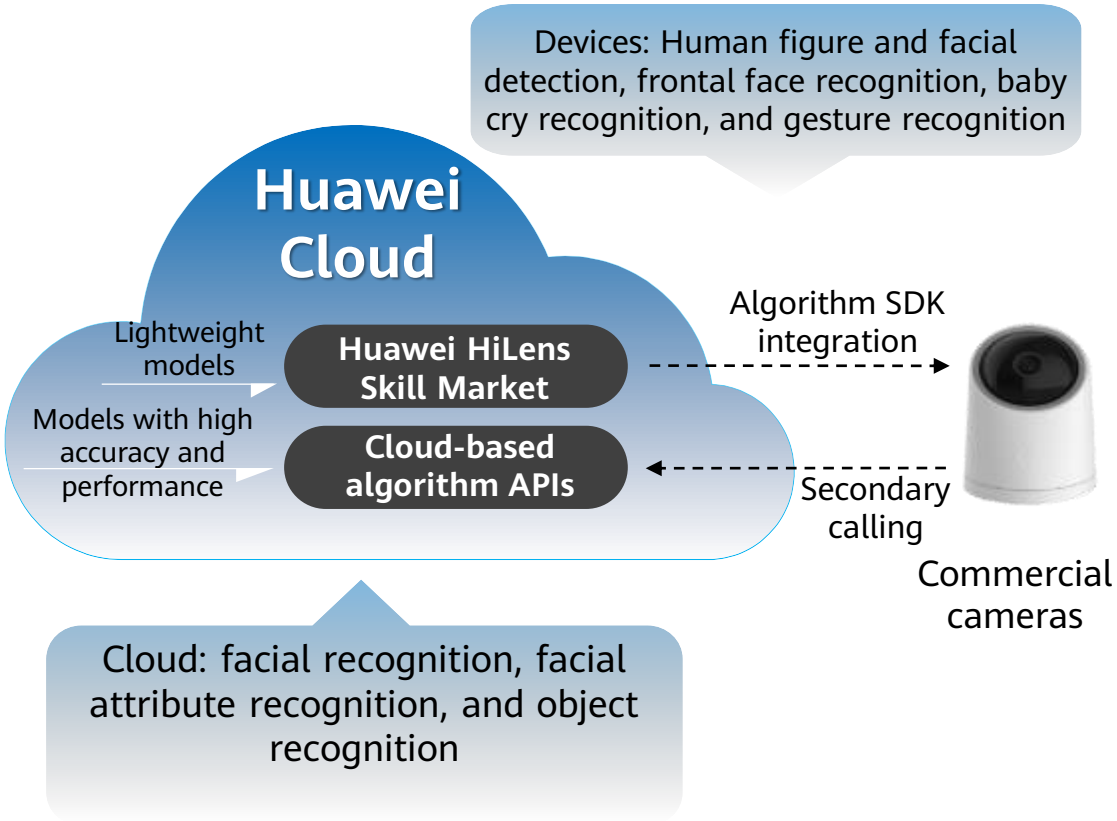
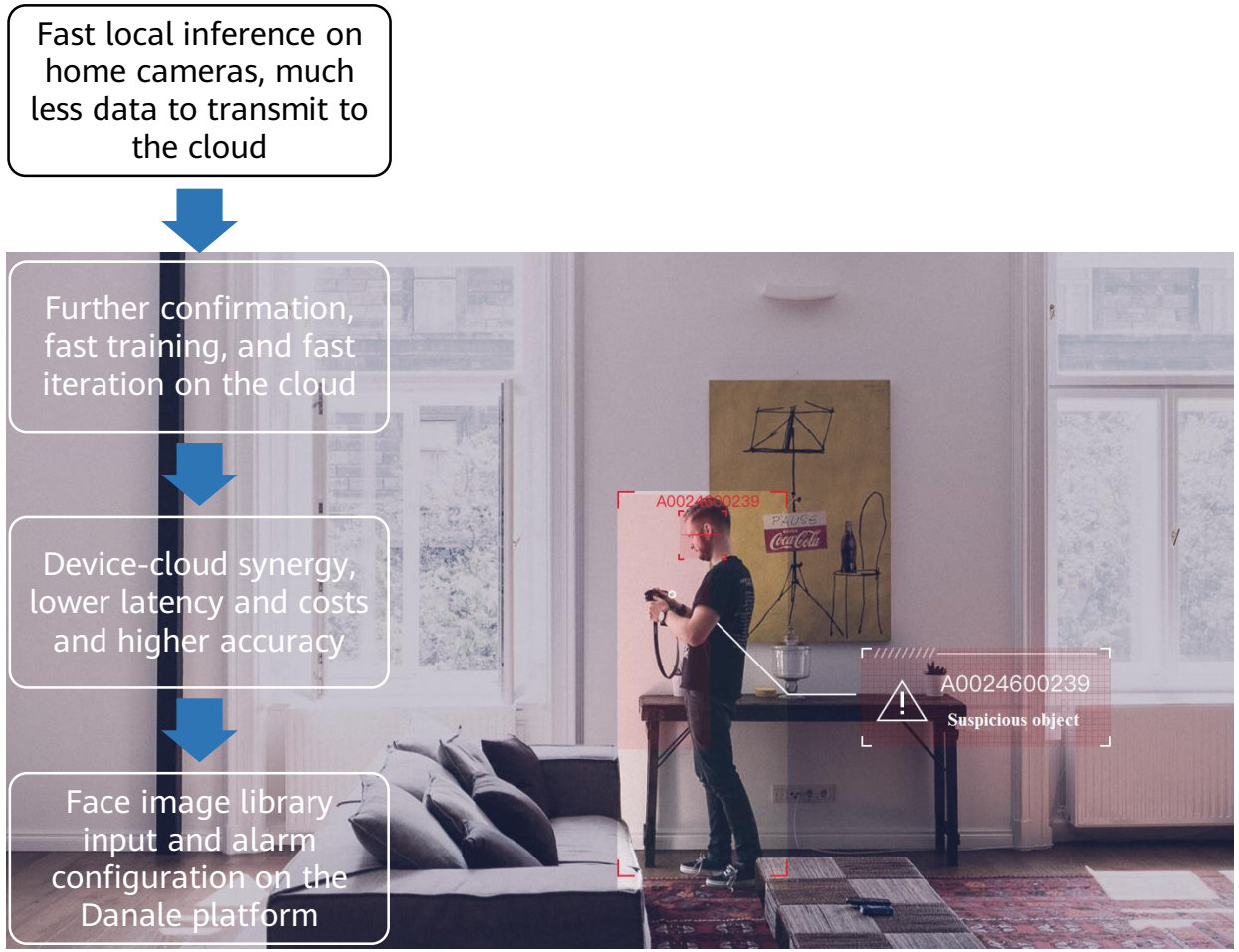
# Danale Smart Cameras for Home Surveillance




Device-cloud synergy: Devices call cloud-based APIs for further confirmation, with results returned in seconds.



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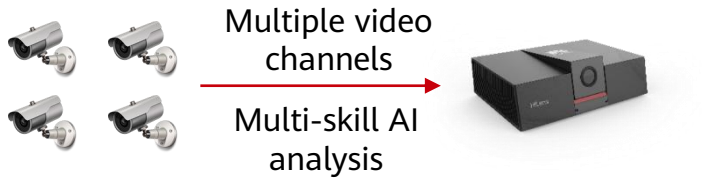
# Huawei HiLens + NTT Data: Smart Auto 4S Store

- NTT Data provides an overall smart automobile 4S store solution that integrates Huawei services.
- Huawei provides a HiLens-based solution to support in-store customer journey and experience analysis and optimization.




One-stop development on cloud and deployment on devices

Hardware differences hidden, one-click skill deployment, and fast iteration

Multi-channel and multi-skill running, at low costs

**Application scenario**

- No personalized precision marketing solutions
- Limited bandwidth, incapable of centralized analysis of all video data
- Different algorithm requirements for different stores

- **Entrance**: customer profile (gender, age group, number of visits to the store, and license plate)
- **Reception area**: VIP identification and customer flow statistics
- **Dashboard area**: recognition of customer interest per car model and emotion
- **Car experiencing**: stay duration and path



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# Zhi Yi Ting + Huawei HiLens: Powering a Hassle-Free Parking Management System with Device-Cloud Synergy

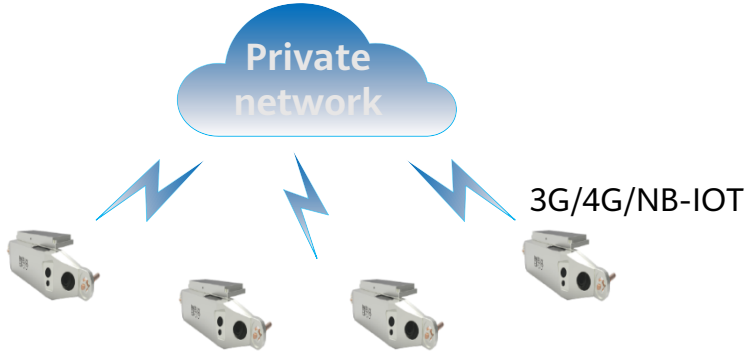
## Intelligent parking service

- Parking management client
- Network server
- Comprehensive management system
- Data statistics and analysis

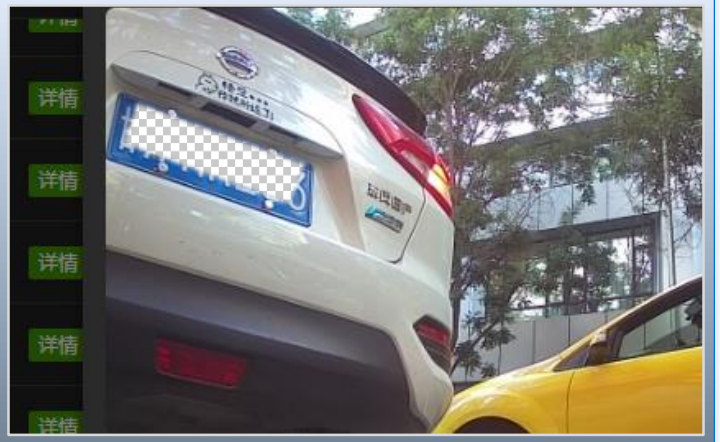
VM & storage

HiLens-based license plate recognition

## Huawei Cloud




License plate capture & quality filtering



Device-cloud synergy, low-position, wide-angle, all-weather license plate recognition

**75%** Recognition accuracy

Automatic locating and navigation to idle parking spaces

**50%** Manual review rate

Camera Automatic day/night mode switchover

**30%** Complaint rate



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# ModelArts & HiLens Device-Cloud Synergy for Unmanned Vehicle Developer Challenge

Traffic light recognition

Driving route recognition

Dynamic obstacle detection

Automated valet parking

Challenges:
Deploying a model on a conventional industrial computer is complex.
Autonomous vehicles run at a high speed, requiring real-time inference.
Algorithms must be able to be updated instantly.

**Data processing**

ModelArts, Huawei's AI development platform, integrates dataset upload, storage, auto labeling, and processing.

**Training on cloud**

ModelArts-trained models are developed into skills through HiLens and delivered to HiLens Kit devices.

**Platform development**

HiLens Kit cameras, USB cameras, network cameras, and Wi-Fi interfaces are provided to facilitate development.

**Built-in framework**

HiLens Framework provides access, data processing, model management, and output interfaces. Developers can focus on algorithm tuning.

**Real-time update**

Skills are updated in real time and delivered through the platform.



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# Section Summary

- This chapter describes the main functions, advantages, and use cases of Huawei HiLens.

# Q&A

1. (Multiple-answer question) Which of the following are advantages of Huawei HiLens?

A. Collaborative inference with device-cloud synergy

B. Unified skill development platform

C. Cross-platform design

D. A rich selection of skills in the skill market



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# Q&A

2. (Multiple-answer question) Which of the following are use cases of Huawei HiLens?

A. Intelligent home monitoring

B. Intelligent campus monitoring

C. Intelligent shopping mall and store monitoring

D. Smart in-vehicle devices



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# Acronyms and Abbreviations

- VR: virtual reality
- AR: augmented reality
- OBS: object storage service
- SDK: software development kit
- OCR: optical character recognition



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# Recommendations

- Huawei Cloud: https://www.huaweicloud.com/en-us/
- HiLens documentation: https://support.huaweicloud.com/en-us/productdesc-hilens/hilens_01_0001.html
- Huawei HiLens official website: https://www.huaweicloud.com/en-us/product/hilens.html
- Huawei HiLens developer community: https://bbs.huaweicloud.com/forum/forum-771-1.html
- HiLens Kit introduction: https://www.huaweicloud.com/product/hilens/hilenskit.html
- Huawei HiLens developer courses: https://education.huaweicloud.com/programs/7515dce4-3f29-4124-a42d-5b1fc11389ce/about
- ModelArts: https://www.huaweicloud.com/product/modelarts.html

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